IRIS Project Guide

Intent and Risk Intelligence System · a guide for the team · Oct 15, 2026

What IRIS is

IRIS is a mental-health support web app for students. A person signs up, completes a short clinical check-in, and can write to IRIS in a chat. **Every message is read by a person on the team**: software never closes a case on its own. When IRIS sees risk it shows helplines straight away, alerts the team, and can offer the person a doctor. A doctor can accept the case, read a summary, talk by chat or video, and record the session.

Status: a working prototype on a local sandbox with fictional accounts. **It is not for real people yet.** No clinician has reviewed the wording, the questionnaire, the session form or the safety rules.

Three principles run through every part:

- **A person reads every message.** The software only sets priority; it never decides a case is fine.
- **Honesty.** IRIS never tells someone that help is coming, or when, before a person has actually accepted.
- **Consent first.** Nothing is shared with a doctor, or with an AI tool, until the person agrees.

The three roles

A role is set only by the backend (never at sign-up), and each role lands on its own pages.

Role	Who	What they do	What they cannot do
Person (user)	A student who needs support	Signs up, consents, does the check-in, chats with IRIS, asks for a doctor (or says yes to an offer), talks to the doctor	See scores, priority labels, reviewer notes, flags or the doctor's clinical note; see anyone else's data

Role	Who	What they do	What they cannot do
Reviewer	A trained team member (counsellor in real use)	Reads every message in a queue sorted by priority, raises priority (never lowers it), records a review, offers a doctor, sees alerts for urgent messages and waiting doctor requests	Lower a priority; chat with the person; share a person's data with a doctor without the person's yes
Doctor	A clinician on the IRIS team	Goes on or off duty, sees open requests, accepts one, reads the summary, starts the conversation, chats or calls by video, ends with a session form	See message text or scores before accepting; see another doctor's cases

Every time a reviewer or doctor opens sensitive data (a record, a queue, a conversation, a summary, a note) it is written to an **audit log**.

Features

Account and onboarding

- Sign up, sign in, forgot and reset password (Supabase Auth, sessions in httpOnly cookies).
- Consent screen; nothing works until it is accepted. Under-18s are recorded with a guardian (not yet verified by email).

Check-in

- PHQ-9 (low mood), GAD-7 (anxiety), daily functioning, and C-SSRS-based self-harm questions.
- A worrying self-harm answer shows helplines **at once** and sets a **safety floor** (none, medium, high) that later raises the priority of every message.
- Due again every 14 days; can be skipped.

Chat with IRIS

- Every message gets a reply and goes to the review queue.
- **Explicit self-harm language** (for example "I'm thinking of killing myself"): an urgent reply, tap-to-call helplines, an **instant alert** for reviewers, and an **offer of a doctor**.

- **Worrying language** (for example "I feel like a burden"): a careful reply with helplines; no tips that could feel dismissive.
- **Ordinary messages**: a short conversational reply with one follow-up question, written by an AI model (optional) and checked before it is shown.
- **Good news and greetings**: a warm, plain reply.
- An honest status line on each message: waiting, being read, or reviewed.

Review queue (reviewer)

- Messages and check-ins sorted by priority; open a record (audited); raise priority with a reason; record a review.
- Live banners: urgent messages not opened yet, urgent doctor requests waiting, requests waiting longer than the goal.
- "Offer a doctor" on a message.

Talk to a doctor

- The person asks from the header or Home, or says yes when IRIS offers (after an explicit message or a High check-in), or when a reviewer offers.
- Honest waiting status with helplines; after 5 minutes with nobody accepting, the person is told plainly and the team is alerted.
- Doctor: on/off duty, open requests (urgent first), accept (first doctor wins), a summary of the person, **Start conversation**, chat, video call, and a required **session form** to close.
- Summary for the doctor: a written "In short" (rules), the check-in, and an optional AI overview the person must agree to separately.

Pages

Page	Route	Who	What it is for
Sign in	`/login`	Everyone	Sign in; sends each role to its own home
Create account	`/register`	New person	Name, email, password, date of birth, phone, emergency contact (guardian if under 18)
Forgot / reset password	`/forgot-password`, `/reset-password`	Everyone	Email link to set a new password

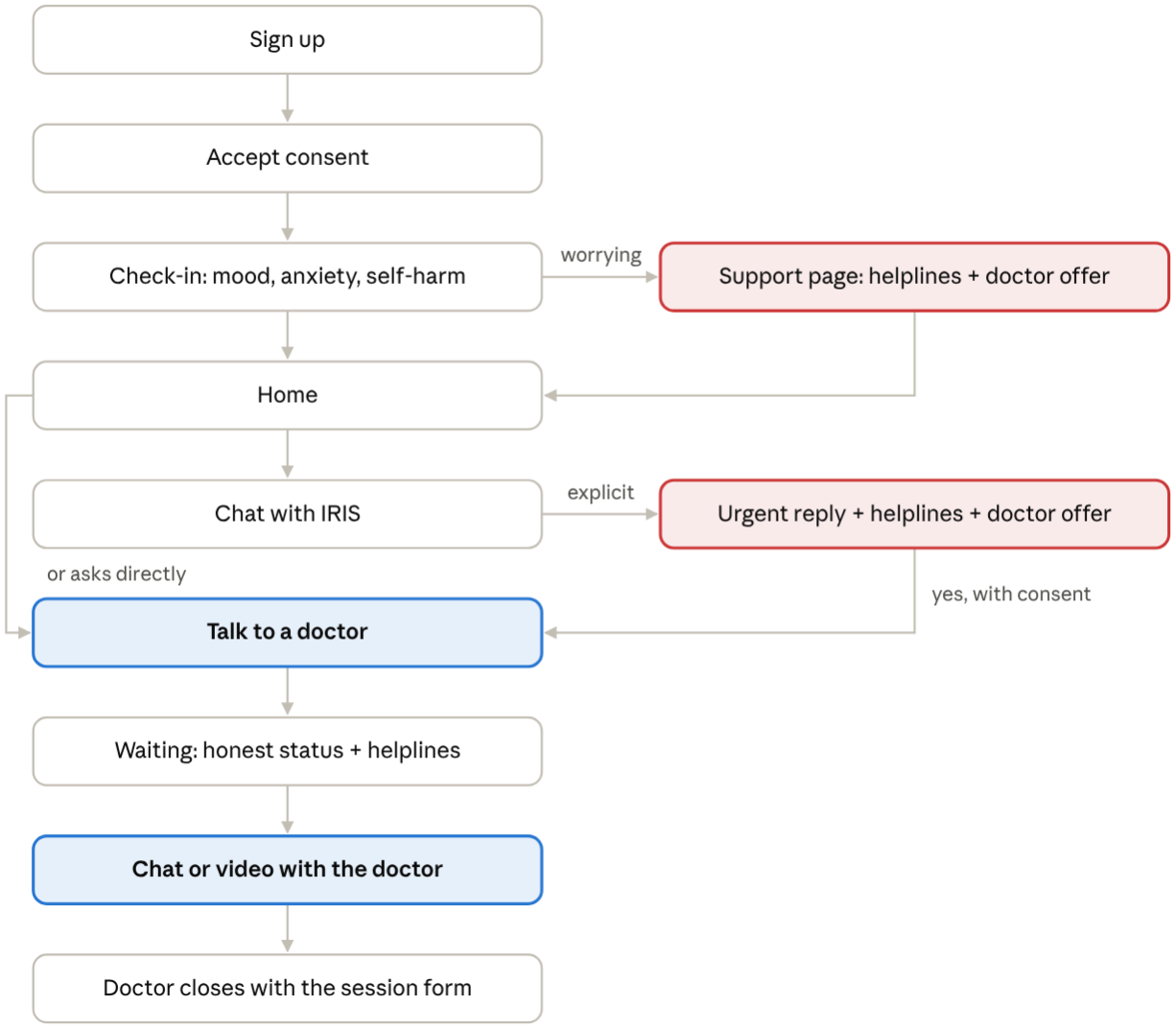
Page	Route	Who	What it is for
Consent	`/consent`	Person	What IRIS is and is not, what is stored; must be accepted first
Check-in	`/check-in`	Person	The clinical questionnaire, one question at a time
Support	`/support`	Person	Shown after a worrying check-in: helplines, and the doctor offer
Home	`/`	Person	Greeting, Talk to IRIS, check-in status, Talk to a doctor (or the doctor offer)
Chat	`/chat`	Person	Talk to IRIS: replies, helplines, status line, the doctor offer
Talk to a doctor	`/talk-to-a-doctor`	Person	Ask for a doctor, see the honest status, then chat and video with the doctor
Reviewer	`/reviewer`	Reviewer	Two tabs, Check-ins and Messages; alerts; open, raise, review, offer a doctor
Doctor home	`/doctor`	Doctor	Duty switch, open requests, Your cases (with a short written summary), closed cases
Case	`/doctor/cases/:id`	Doctor	Conversation in the centre; the person's card and summary on the right; Start, video, chat, End session form

A crisis strip ("In crisis or need to talk now? Call Tele-MANAS 14416") is on every page a person sees, and the footer always says IRIS is not an emergency service.

Flow charts

1. The person's journey

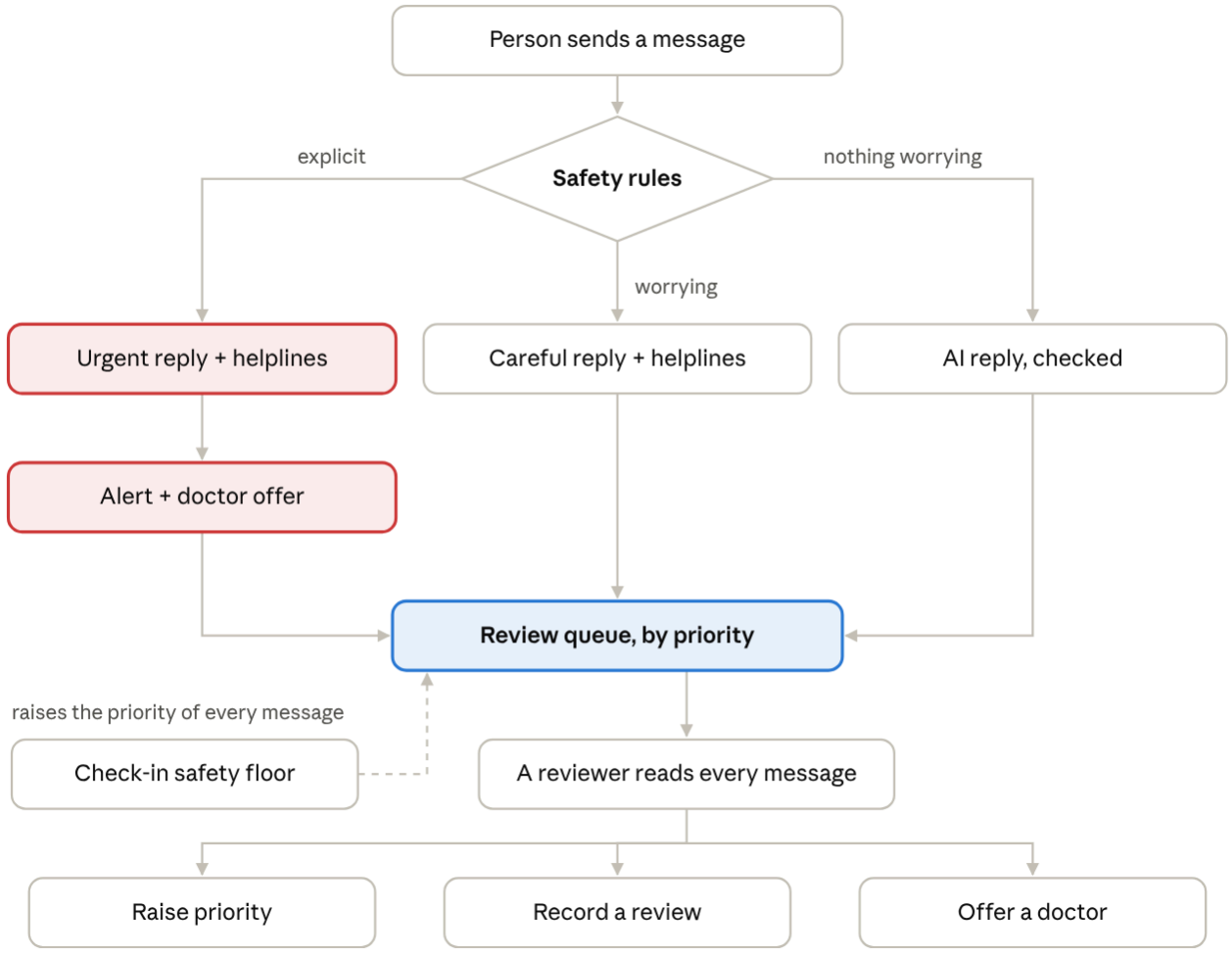
From sign-up to talking with a doctor. Helplines are on screen at every step where risk shows up.



The person's journey · sign-up to the doctor

2. How one chat message is handled

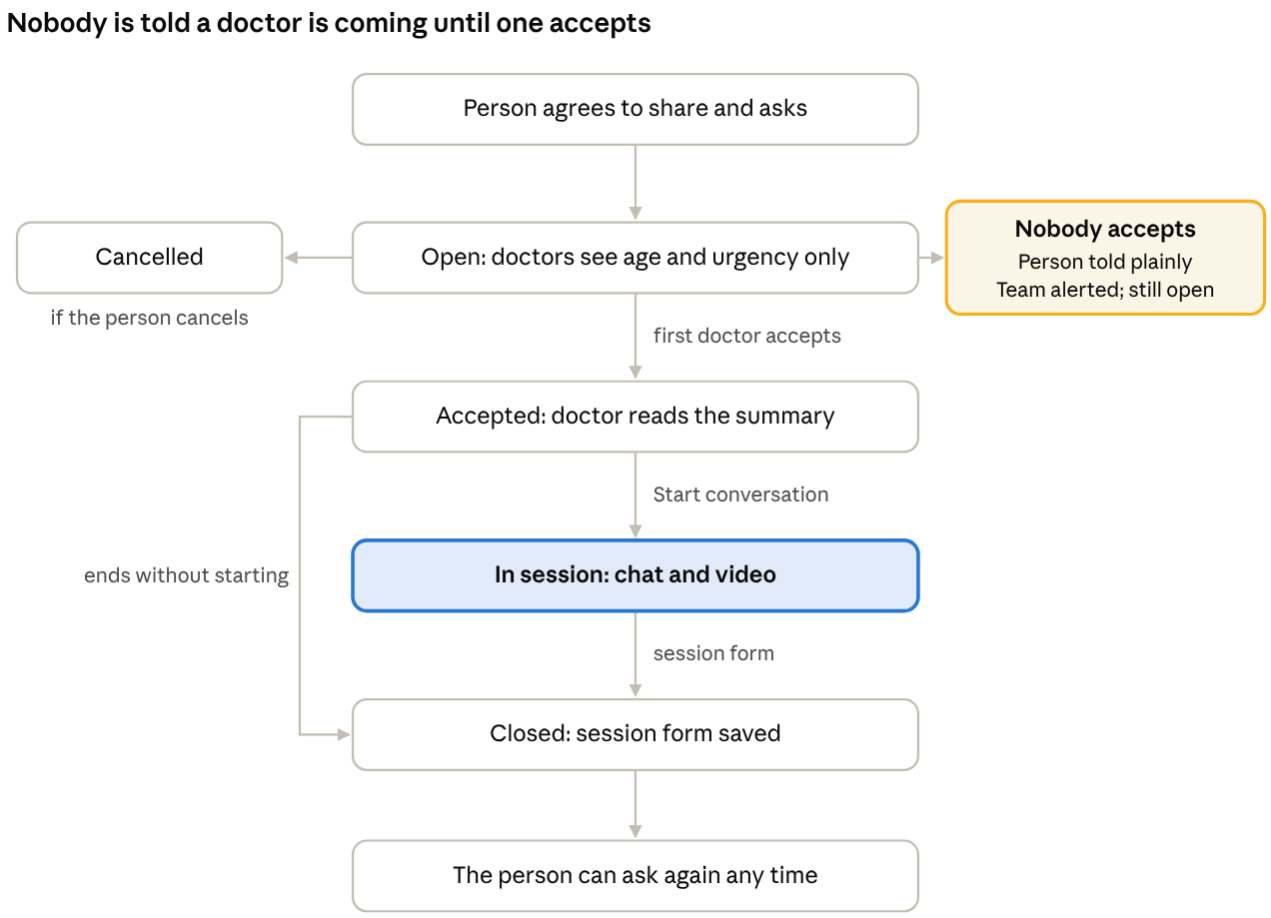
The rules and the check-in floor set the priority. A person reads every message, whatever the priority.



How one chat message is handled

3. A doctor request, start to finish

The states a request moves through. Before a doctor accepts, IRIS never says one is coming.



A doctor request, start to finish

How priority and safety are decided

No machine-learning model decides anything about risk. Priority comes from two things only:

1. **Safety rules** that look for explicit and indirect self-harm language (for example "kill myself", "want to die", "take all my pills", "I feel like a burden"). They tolerate masked spellings such as "k*ll" and "su1cide", and ignore everyday phrases such as "this homework is killing me".
2. **The check-in safety floor** from the self-harm questions. A High floor makes every message from that person High priority, but a calm message from them still gets a calm reply.

The highest of the two wins, and nothing can lower it. Then a person reads the message.

Why no model? In Phase 2 the team tested 21 model set-ups and a fine-tuned DistilRoBERTa. None reached the target (finding 95% of high-risk cases while reviewing 30% of messages), and they raised many false alarms on ordinary chat. So the safe design is rules plus a human read of everything.

Where AI is used (all optional, all checked):

Where	What it does	Guard rails
Chat reply	A short reply with one follow-up question for **ordinary** messages only	Never used for worrying or urgent messages; output checked for promises, diagnoses, times, numbers; falls back to fixed wording
Doctor's overview	A short paragraph about what the person wrote	Needs the person's separate tick; sends no name or contact details; reassuring sentences ("no risk factors") are removed; labelled as a draft

The written "In short" summary and all counts are built by plain rules and never leave the server.

Safety and privacy rules

- **Helplines always visible** where risk appears: Tele-MANAS 14416, KIRAN 1800-599-0019, emergency 112, all tap-to-call.
- **No false promises.** Before a doctor accepts, the person reads "we cannot say how long". The 5-minute wait is a goal for the team only and is never shown to the person. Tests check this wording.
- **Consent before sharing.** A doctor request needs the person's tick. "Not now" shares nothing. A reviewer can only *offer* a doctor.
- **Minimum before accepting.** On-duty doctors see first name, age, urgency, the reason and the check-in level. They see message text, scores and the summary only after accepting.
- **One doctor per case.** Accepting is atomic; a second doctor is refused.
- **The person never sees** scores, priority labels, worrying-language flags or the clinical note.
- **Under 18** is flagged to the doctor with the guardian's details.
- **Audit log** for every opening of sensitive data and every key action.
- **Video is direct** between the two browsers; the server only passes set-up messages and deletes them when the call ends. Nothing is recorded.
- **Roles are set only by the backend**, never chosen at sign-up.

Tech stack and code layout

Layer	Stack
Frontend	React 19, TypeScript, Vite, Tailwind CSS 4, React Router
Backend	FastAPI, SQLAlchemy 2, Alembic migrations, Pydantic 2
Database	Supabase Postgres with Row Level Security on every table
Auth	Supabase Auth; tokens in httpOnly cookies
Safety rules and evaluation	Python package in `ml/` (rules, decision policy, reply wording, test suites)
Optional AI	Groq (`openai/gpt-oss-120b`), off unless a key is configured
Video	WebRTC, peer to peer, with a public STUN server

```
backend/app/
  routers/      auth, check-in, messages (chat), reviewer, consults (person's
doctor requests), doctor
  services/     scoring, safety, triage (rules + floor), companion (AI chat
reply),
                consult (doctor requests), case_summary (doctor's summary)
  models.py, schemas.py
backend/migrations/   0001 to 0006
backend/tests/        226 tests
frontend/src/pages/   Login, Register, Consent, CheckIn, Support, Home, Chat,
AskDoctor,
                      Reviewer, ReviewerMessages, Doctor, DoctorCaseView
ml/                   safety rules, decision policy, reply wording, clinical
review pack, Phase 1-2 reports
docs/                 PROJECT_SUMMARY, DOCTOR_FEATURE, ARCHITECTURE,
TRIAGE_INTEGRATION_SPEC, DEMO_ACCOUNTS
```

How to run it

The code is on GitHub: https://github.com/JathinSekhar/IRIS

Local sandbox (no Supabase needed, fake sign-in, throwaway database):

```
cd backend && python -m devtools.demo_server
cd frontend && npm install && npm run dev
```

Then open http://localhost:5173. The sandbox creates a reviewer and a doctor on start; patients sign up through the app. Login details for the demo accounts are in `docs/DEMO_ACCOUNTS.md` in the repository, which is generated from the code so it is always up to date. All accounts are fictional; never use real people's details.

Tests:

```
cd backend && pytest
cd ml && pytest
cd frontend && npm run build
```

A short demo to show the team:

1. Sign in as a **low-risk** person and send "I'm stressed about my exams". A friendly reply, no doctor offer.
2. Sign in as a **high-risk** person. Home shows "Would you like to talk to a doctor?". Tick and say yes.
3. Sign in as the **doctor**. Turn on duty, accept the request (marked Urgent, High check-in), read the summary, Start conversation.
4. Sign in as the **reviewer** to see the queue and the alerts.

Status, gaps and next steps

What is tested: 226 backend tests and 589 ML tests pass. A 58-scenario wording check shows 43 behaving as intended, 13 that need a clinician's decision, and 2 known gaps. These show the software does what it is designed to do.

What is not measured: how well IRIS finds real risk. That needs a gold set labelled by two independent annotators with a counsellor adjudicating. **We make no accuracy or safety claim until then.**

Known gaps:

- Hinglish, slang and indirect phrasing are often missed by the rules.
- No clinician has reviewed the wording, the questionnaire or the session form.
- The consent text does not yet mention the doctor feature or the optional AI.
- Database migrations 0002 to 0006 are not applied to Supabase; the doctor feature has only run in the sandbox.
- Message text is not encrypted and has no retention period.
- Video has no TURN relay, so it can fail on strict networks (the chat still works).

- Real use needs doctor verification, staffing, and a privacy review of every third party (Groq, the STUN server).

Next steps:

1. Clinician review of all wording, the questionnaire and the session form.
2. Update the consent text, then apply the migrations to Supabase.
3. Build the gold set and run one honest evaluation.
4. Improve detection of Hinglish and slang.
5. Encryption, a retention period, and a TURN server before any real pilot.